Active Learning through the Graph Transduction Game

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Abstract

Deep networks are data-hungry, yet high-quality annotation is expensive, which renews the case for active learning (AL): selecting the few samples whose labels help the most. We revisit AL through the lens of the *Graph Transduction Game* (GTG), an evolutionary, non-cooperative game whose replicator dynamics propagate labels from a small labelled set to unlabelled points and settle at a consistent (Nash) labelling. Our key observation is that the *trajectory* of a sample's label entropy under these dynamics is a strong signal of informativeness: points that remain uncertain or oscillate the longest are the most valuable to annotate. We introduce **ALGTG**, an acquisition function built on this signal. We further show that avoiding redundant queries need not be hard-coded into the payoff: once labelled players are clamped, large parts of the payoff matrix provably do not influence the game, and the suppression of samples already well explained by the labelled set *emerges* from the dynamics themselves. We make this precise and add a single-parameter term that amplifies the effect where it is actually effective. Finally, we present an end-to-end differentiable variant that learns the signal jointly with the backbone. On CIFAR-10/100, SVHN, Fashion-MNIST and Tiny-ImageNet, ALGTG

1 Introduction

Modern deep networks owe much of their success to large annotated corpora, but annotation is the one ingredient that does not become cheaper with scale. In specialised domains—medical imaging, remote sensing, speech—labels require expert time, and the marginal value of an additional label varies enormously from sample to sample. Active learning (AL) addresses exactly this asymmetry: given a fixed annotation budget, choose the samples whose labels will improve the model the most (Settles, 2009).

Most deep AL methods answer that question with a *static* score. Uncertainty methods read a scalar off the current model—predictive entropy, margin, or the mutual information between prediction and parameters (Houlsby et al., 2011; Gal et al., 2017). Diversity methods ignore the model's confidence and instead cover the representation geometrically (Sener and Savarese, 2018). Hybrids combine the two cues in a single embedding or objective (Ash et al., 2020; Parvaneh et al., 2022), and a further line trains an auxiliary head to predict the signal (Yoo and Kweon, 2019; Kim et al., 2021; Kye et al., 2023). What these approaches share is that the score is a snapshot: it is computed from a frozen model at one instant, and it says nothing about how a hypothetical label assignment would *behave* once it had to be reconciled with the labels of every other sample.

Transductive learning offers precisely that missing view. In the Graph Transduction Game (GTG) of Erdem and Pelillo (2012), labelling is cast as a non-cooperative game: samples are players,

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labels are strategies, and a payoff rewards agreeing with similar players. The replicator dynamics—a continuous generalisation of relaxation labelling (Pelillo, 1997)—propagate the labelled seeds through the data and settle at a Nash equilibrium, a labelling consistent with both the supervision and the manifold structure. Existing work uses the *outcome* of this process: the equilibrium labelling. Our starting point is that the *process itself* is informative, and that for active learning it is in fact the more useful object.

The reason is intuitive. A sample lying deep inside a region the labelled set already explains receives a strong, coherent pull toward one label and commits after a couple of iterations. A sample on a decision boundary receives conflicting pulls and stays ambivalent; a sample in a region no label reaches receives almost no pull at all and remains undecided. Tracking each sample's label entropy *across* the iterations therefore yields a single quantity in which uncertainty and redundancy are already entangled the way an acquisition function needs them to be—without hand-combining two heuristics through a trade-off hyperparameter. This is the signal ALGTG queries on.

Making this precise required understanding which parts of the game actually matter. Because labelled players are clamped to their known labels, we show (Proposition 1) that entire blocks of the payoff matrix cannot influence the outcome: the natural-looking idea of making already-annotated samples "repel" each other, so as to discourage redundant queries, provably does nothing. The redundancy avoidance that such a term is meant to produce is instead an *emergent* property of the dynamics, and we give a one-parameter term that modulates it where it is genuinely effective. Finally, since the whole procedure is differentiable, we derive an end-to-end variant in which the acquisition signal is learned jointly with the backbone, letting representation and selection co-adapt.

Contributions.

- A novel AL acquisition function based on the *entropy trajectory* of samples under Graph Transduction Game dynamics.
- An analysis of *which blocks of the payoff matrix actually drive the game* once labelled players are clamped (Proposition 1). It follows that redundancy avoidance is an emergent property of GTG rather than something that must be engineered into the payoff; we also derive a one-parameter term that amplifies it where it is effective.
- An end-to-end differentiable variant that learns the acquisition signal jointly with the task network, so representation and selection co-adapt.
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2 Related Work

Uncertainty-based AL. Classical strategies query the points where the current model is least confident, using predictive entropy, margin or least confidence (Settles, 2009). Bayesian variants estimate epistemic uncertainty via the mutual information between predictions and model parameters (BALD) (Houlsby et al., 2011), made practical for deep nets through MC-dropout (Gal et al., 2017) and extended to the batch setting by BatchBALD (Kirsch et al., 2019). Pure uncertainty ignores redundancy and tends to select correlated batches.

Diversity and hybrid AL. Representation-driven methods instead cover the feature space: Core-Set casts batch selection as a k-center problem over embeddings (Sener and Savarese, 2018). Hybrid methods combine both cues: BADGE runs k-means++ on *gradient* embeddings, whose magnitude encodes uncertainty and direction encodes diversity (Ash et al., 2020); AlphaMix probes

for decision-boundary instability by interpolating unlabelled features toward labelled class anchors (Parvaneh et al., 2022); CDAL measures contextual diversity through KL divergence of predictive distributions (Agarwal et al., 2020).

Learned acquisition. A parallel line learns the acquisition signal. Learning Loss attaches a module that predicts each sample's loss and queries the highest-loss points (Yoo and Kweon, 2019). VAAL trains a VAE–discriminator to tell labelled from unlabelled in latent space (Sinha et al., 2019), and TA-VAAL makes it task-aware by feeding in a learned loss/rank (Kim et al., 2021). TiDAL exploits training dynamics as a supervisory signal for the acquisition head (Kye et al., 2023).

Graph-based and transductive AL. Closest to our work, Sequential GCN builds a graph over labelled and unlabelled samples and learns node representations to rank informativeness (Caramalau et al., 2021). Our method is also graph-based but differs fundamentally: rather than learning graph embeddings, we run the *Graph Transduction Game* (Erdem and Pelillo, 2012)—an evolutionary game whose replicator dynamics generalize classical relaxation labelling (Pelillo, 1997)—and derive the acquisition signal from the *temporal evolution* of label entropy under those dynamics. To our knowledge, GTG dynamics have not previously been used as an AL acquisition function.

3 Background: The Graph Transduction Game

Given n objects (players) and c labels (pure strategies), GTG casts transductive labelling as a non-cooperative game (Erdem and Pelillo, 2012). Player i holds a mixed strategy $x_i \in \Delta^{c-1}$ (a distribution over labels), and players receive payoff for agreeing with *similar* players, encoded by a non-negative affinity matrix $W \in \mathbb{R}^{n\times n}$. Writing $x = (x_1, \ldots, x_n)$, the population evolves by the multi-population replicator dynamics

$$x_i^{(t+1)} \;=\; x_i^{(t)} \odot \frac{\left(Wx^{(t)}\right)_i}{x_i^{(t)\top}\left(Wx^{(t)}\right)_i}, \tag{1}$$

where $\odot$ is the elementwise product and the ratio is taken per label. Labelled players are clamped to their one-hot strategies throughout. The dynamics increase average payoff monotonically and converge to a Nash equilibrium—a labelling that is consistent with the affinity structure and the labelled seeds.

4 Method: ALGTG

Let L and U be the labelled and unlabelled sets and $\phi(\cdot)$ the backbone feature map. For a pool of $n = |L| + |U|$ samples we build a payoff matrix, run the game, and score unlabelled samples by their entropy trajectory.

Payoff construction. We form W from a similarity kernel over features, $W_{ij} = k(\phi(x_i), \phi(x_j))$, with k one of cosine similarity, Pearson correlation, or a self-tuning RBF kernel whose per-point bandwidth is the distance to the 7th nearest neighbour (Zelnik-Manor and Perona, 2004). Because the replicator update (1) requires a non-negative payoff to keep x_i on the simplex, we clamp W to $[0, 1]_{\geq 0}$ and use a zero diagonal, so no player is rewarded for agreeing with itself. An optional threshold (a fixed value or the matrix mean) sparsifies weak edges.

Initialization and dynamics. We set $x_i^{(0)} = e_{y_i}$ for $i \in L$ and $x_i^{(0)} = \frac{1}{c}\mathbf{1}$ for $i \in U$, then iterate (1) re-clamping the labelled rows each step, until $\|x^{(t+1)} - x^{(t)}\| < \varepsilon$ or a maximum of T iterations. If a player receives no payoff from anyone—an isolated node, which sparsification can create—its update is $0/0$; we adopt the natural convention that such a player has no incentive to deviate and keep $x_i^{(t+1)} = x_i^{(t)}$. This keeps every player on the simplex; normalizing by a floored denominator would instead collapse the row to the zero vector, silently removing the point from the game and assigning it zero entropy.

Which blocks of the payoff drive the game? Partitioning W into blocks $W_{LL}, W_{LU}, W_{UL}, W_{UU}$ induced by L and U yields a simple but consequential observation.

Proposition 1. *If the labelled players are clamped at every iteration, then the trajectory $\{x_U^{(t)}\}_{t\geq 0}$ of the unlabelled players—and hence any acquisition score derived from it—depends on W only through W_{UU} and W_{UL}. The blocks W_{LL} and W_{LU} are inert.*

Proof sketch. The update of player i in (1) depends on W only through row i, namely $(Wx)_i = \sum_j W_{ij}x_j$. For $i \in U$ this row meets only the columns of L and U within W_{UL} and W_{UU}. Rows indexed by $i \in L$—the only rows in which W_{LL} and W_{LU} appear—are overwritten by the clamp before they are used. □

Proposition 1 has a practical corollary: any attempt to steer the acquisition by editing W_{LL} (for instance, turning labelled–labelled affinities into distances so that annotated points "repel" each other, an intuitive design we initially adopted) has *no effect whatsoever* on the selected batch. We verify this numerically: such edits leave the acquisition scores identical to machine precision.

Redundancy avoidance is emergent. Rather than a defect, this delimits where the mechanism actually lives. An unlabelled point lying close to the labelled set receives, through W_{UL}, a strong and coherent pull toward a single label; it commits almost immediately, so its entropy trajectory is short and its score low. A point in a region the labelled set does not explain receives weak or conflicting pull and remains uncertain for many iterations, earning a high score. Standard GTG dynamics therefore already suppress redundant queries, with no bespoke payoff term. To *modulate* the strength of this effect we introduce a single non-negative parameter λ that scales the labelled–unlabelled coupling,

$$W_{UL} \leftarrow (1+\lambda)\, W_{UL}, \qquad W_{LU} \leftarrow (1+\lambda)\, W_{LU}, \tag{2}$$

with $\lambda = 0$ recovering standard GTG. Because clamped labelled players enter the potential of the restricted game on U only through a linear term, (2) preserves non-negativity and the symmetry of W_{UU}, so the Baum–Eagon convergence guarantee is unaffected. Empirically, increasing λ monotonically widens the score gap between redundant and informative samples.

Acquisition by entropy trajectory. For each unlabelled sample we record the Shannon entropy of its strategy at every iteration, $H_i^{(t)} = -\sum_k x_{ik}^{(t)} \log x_{ik}^{(t)}$, and aggregate the trajectory either by its mean or by its area (trapezoidal integral):

$$s_i \;=\; \tfrac{1}{T}\sum_{t=1}^{T} H_i^{(t)} \quad \text{or} \quad s_i \;=\; \textstyle\int_1^T H_i^{(t)}\, dt. \tag{3}$$

We query the top-k unlabelled samples by s_i. Intuitively, a point whose label assignment settles immediately is easily explained by its neighbours and carries little new information, whereas a point

Algorithm 1 ALGTG acquisition (offline)

Require: features $\phi(x)$ for $L \cup U$; budget k; iterations T; tolerance ε

build the payoff W (kernel, clamp to $[0,1]_{\geq 0}$, zero diagonal, threshold)
initialise $x^{(0)}$: one-hot for $i \in L$, uniform for $i \in U$
for $t = 0, \ldots, T-1$ **do**
    update $x^{(t+1)}$ by Eq. (1); re-clamp the labelled rows
    record $H_i^{(t+1)}$ for $i \in U$
    if $\|x^{(t+1)} - x^{(t)}\| < \varepsilon$ **then break**
end for
$s_i \leftarrow \text{aggregate}(H_i^{(1:T)})$ for $i \in U$
return the top-k unlabelled samples ranked by s_i

that stays uncertain or oscillates under propagation lies where the current model and manifold structure disagree—exactly where a label is most useful.

End-to-end differentiable variant. The offline scheme runs GTG on frozen embeddings after training. We also propose an online variant in which a lightweight module projects intermediate backbone features to a low-dimensional space, runs the *differentiable* dynamics (1) on the induced affinity, and reads the entropy trajectory through an MLP or LSTM head to predict s_i. The head is trained jointly with the task loss by regressing the GTG entropy computed on the full embeddings (a teacher signal), so gradients from the acquisition objective also shape the representation. This mirrors the auxiliary training of Learning Loss (Yoo and Kweon, 2019) but with a game-theoretic, transduction-derived target.

5 Experiments

Datasets. We evaluate on five image-classification benchmarks spanning a range of label cardinalities and image statistics: CIFAR-10 and CIFAR-100 (50k train / 10k test, 32×32), SVHN (73k / 26k, 32×32), Fashion-MNIST (60k / 10k, 28×28, greyscale) and Tiny-ImageNet (100k / 10k, 64×64, 200 classes).

Protocol. Every method starts from the same randomly drawn set of 1000 labelled samples and performs 10 acquisition cycles of 1000 labels each, ending at 11k labels. At each cycle the candidate pool is a fresh random subset of 10,000 unlabelled samples, which bounds the cost of the pool-based methods (ours included) and is applied identically to every baseline. After each acquisition the network is *re-initialised from the same checkpoint and retrained from scratch* for 200 epochs, so that the reported accuracy reflects the acquired labels and not the optimisation history. Each configuration is repeated over 5 trials with different seeds, which also re-draw the initial labelled set; we report the mean and a 95% confidence interval.

Implementation. The backbone is ResNet-18 (He et al., 2016) (VGG-16-bn for Caltech-256), trained with SGD (lr 0.1, momentum 0.9, weight decay 5×10^{-4}, step decay at epoch 160) and batch size 128. The learnable acquisition modules use Adam (lr 10^{-3}). Unless stated otherwise AL-

Table 1: Test accuracy (%) vs. labelling budget. *Placeholder — run* `paper/make_results.py` *to populate.*

Method	2k	5k	10k
Random	–	–	–
...	–	–	–
ALGTG (ours)	–	–	–

GTG uses the correlation kernel, mean-thresholded sparsification, mean aggregation of the entropy trajectory, $T = 30$ iterations, tolerance $\varepsilon = 10^{-4}$ and $\lambda = 0$.

Metrics. Alongside accuracy at each budget we report the area under the budget–accuracy curve (AUBC), normalised by the budget range, which summarises a method's behaviour over the whole annotation trajectory in a single number and avoids conclusions that hinge on one cherry-picked budget.

Reproducibility. All methods share the same backbone, optimiser, schedule, seeds and candidate pools, and differ only in the acquisition function, so the comparison isolates the selection strategy. Every baseline is re-implemented in the same framework rather than copied from published tables. We flag one consequence of this explicitly: while preparing the study we found defects in three of our own baseline implementations—MC-dropout that produced no stochasticity (so BALD's mutual information was identically zero), per-class anchoring that silently fell back to a global mean in AlphaMix, and BADGE clustering penultimate features instead of gradient embeddings—each of which had made the corresponding baseline weaker than it should be. All reported numbers were produced after these were fixed; no pre-fix result is included. Code, configurations and the scripts that turn the raw logs into the tables and figures of this paper are released with it.

Baselines. Random, Entropy, Core-Set (Sener and Savarese, 2018), BADGE (Ash et al., 2020), BALD (Gal et al., 2017), Learning Loss (Yoo and Kweon, 2019), VAAL/TA-VAAL (Sinha et al., 2019; Kim et al., 2021), CDAL (Agarwal et al., 2020), AlphaMix (Parvaneh et al., 2022), TiDAL (Kye et al., 2023), and both variants of the graph-based Sequential GCN —UncertainGCN and CoreGCN—(Caramalau et al., 2021), which are the closest competitors to our setting.

Ablations. (i) affinity kernel (cosine / correlation / RBF); (ii) entropy aggregation (mean vs. integral); (iii) offline vs. online, and head architecture (MLP vs. LSTM); (iv) the redundancy weight λ of Eq. (2), swept from 0 (standard GTG) upward, quantifying how much explicit amplification adds over the emergent behaviour; (v) **the central control**: the acquisition signal itself—the full entropy *trajectory* versus only the entropy at convergence, versus the plain softmax entropy of the backbone. This isolates whether the *dynamics*, and not merely the final labelling or the model's own confidence, carry the information.

Analysis. non-queried samples; t-SNE of selections vs. competitors; cost/runtime discussion.

6 Conclusion

We presented ALGTG, an active-learning acquisition function that scores unlabelled samples by the *trajectory* of their label entropy under the replicator dynamics of a graph transduction game, rather than by a static readout of a frozen model. Along the way we characterised which parts of the game's payoff can influence the outcome at all (Proposition 1), which shows that the avoidance of redundant queries is emergent rather than something to be engineered, and we gave a differentiable formulation that lets the acquisition signal shape the representation it is computed from.

Limitations. Three are worth stating plainly. First, *cost*: the payoff matrix is quadratic in the size of the candidate pool, so the offline variant becomes expensive as the labelled set grows, and the online variant unrolls the dynamics inside the autograd graph. Sub-sampling the pool, as we do, bounds this but makes the transduction depend on the sample drawn. Second, *locality of the online variant*: it runs the game over a mini-batch rather than the full pool, so its equilibrium—and hence its signal—depends on batch composition; it is best understood as an auxiliary objective that shapes the backbone, in the spirit of Yoo and Kweon (2019), rather than as a cheaper substitute for the offline computation, whose teacher signal is available at query time anyway. Third, *graph construction*: the acquisition inherits whatever the affinity kernel and its sparsification encode, and aggressive thresholding can disconnect points from the graph entirely, at which point the dynamics leave them untouched. We keep such nodes on the simplex by convention, but a principled treatment of disconnected candidates—which are often precisely the novel ones—is left to future work.

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